

Playing blackjack

Portfolio assignment 4

MAL2 Autumn 2025

In this assignment, you are tasked with solving the problem of training a model to effectively play the classic casino game blackjack using reinforcement learning. The environment for simulation is provided by Gym, which you can learn more about here:

https://gymnasium.farama.org/environments/toy_text/blackjack/.

Blackjack's simplicity, with only two actions and short game durations, makes it an ideal candidate for rapid training on standard computers. This allows you to experiment with different policies, training methodologies, neural network architectures, and more.

You will implement a basic policy where the player sticks whenever their card value is 17 or higher, which should give you a win rate of approximately 39 %. Then train a neural network to play blackjack and surpass the 39 % win rate achieved by the basic policy. Simulate 1,000 games and report the win rate of your model. If your model wins more often than it loses, you might want to board the next plane to Las Vegas.

You are to hand in a notebook with

- output (all cells must be run)
- relevant comments describing your approach, experiments, and findings
- your basic policy and neural network implementation as well as their win rates
- at least two interesting figures or animations with relevance to the assignment